Machine Learning tools Comparative Analysis assignment - SBA25255

Organization Description

Stripe is a ‘fintech’ which over the years has evolved from being a payments provider to offering a full blown financial infrastructure, Their financial infrastructure manages a full range of services including billing, tax, fraud management, card issuance, identity management and much more. Their mission is to “Grow the GDP of the internet”[1]

Some interesting facts about Stripe -

1. 1.5% of global GDP runs on Stripe[2]
2. 135+ Payment methods are supported including crypto stablecoins[2]
3. Processed $1.4tn worth of payments in 2024[2]
4. 200m+ active billing subscriptions[2]
5. 4.7m websites use Stripe[5]

Given the above scale AI is critical to both Stripe’s and their customers ongoing success and growth.

Stripe’s AI Use case for this assignment

Some Context

Stripe has an automated onboarding process for new customers. Part of this process involves ‘Know your Customer” (KYC)[3] anti money laundering (AML)[4] regulatory requirements.

As part of onboarding the customer is initially asked to key in some specific information about their business. The KYC regulatory requirements for the geographical region the business is located in are then executed. This portion of the process is highly automated and completes within seconds.

However if the above KYC checks fail then the customer is requested to upload identify documentation. The process itself can take hours to complete and involves a human manually inspecting these documents vs the details the customer entered and making a determination as to whether (a) the identification document provided is valid (b) they are who they say they are. The details of each identity document review are recorded in an online system.

Abstraction -Attributes of the problem to be solved

- When a customer submits an identity document we want to ***predict*** -
  - The document itself is valid
  - The information entered by the customer matches what's on the identification document

- The above requires a series of ***decisions*** to be made. For example -
  - Does the format of the document match how it should appear? For example if it's a french drivers licence the expiry date should be on the front on line 4b
  - Is the expiry date < today ?
  - Does the information entered by the customer match what's on the document

- We have a system that records details of all of the manual reviews. I.e. we have ***labelled*** data.

- Stripe has had year on year (yoy) growth of 38%[4] so we can assume this is a **high volume / large data set environment.**

We can use the above attributes to identify the AI solution that best fits the problem we need to solve

Potential Approaches

Unsupervised or supervised learning?

We want to predict upon document submission that the document itself is valid and that the customer is who they say they are. We want to leverage historical labelled data from previous document reviews. This is strongly aligned with supervised learning.

Unsupervised learning is best used for gaining insights into patterns and structure of data where a “correct answer” is not required. So unsupervised learning will not solve our problem.

Therefore supervised learning best fits the problem we need to solve.

Classification or clustering ?

Clustering is grouping data points into clusters and then reviewing the clusters to gain insights. It’s pure data discovery where there is no explicit answer or prediction.

Classification is the opposite to clustering in that it does involve providing an explicit prediction. It also uses labeled data as an input to making the prediction. Classification can predict either a category such as determining whether a given email is spam. Or predict a continuous value such as what the outside temperature will be tomorrow.

Our use case needs to predict an identity verification outcome using historical labelled data as an input. Therefore a classification solution that will predict a category (as opposed to a continuous value) is most aligned to our solution.

Algorithm options

KNN -

This algorithm is for unsupervised learning. It clusters data when in turn provides insights on data structure and patterns. It predicts a ;label of a data point from labels of closest neighbours. There's no training required and it works well in small data sets.

However, given we have decided to choose supervised learning KNN is not applicable for our solution. In addition to this our use case involves large datasets which would also be another reason to rule out this option.

Naive Bayes

This algorithm is for supervised learning and has the following attributes -

- It assumes features are independent of each other
- It's used as a classifier for predicting categories
- It's fast and scalable
- Works well with high dimensioned text based data

All of the above fit the solution we are looking for. However there is a gap with regard to the algorithm being capable of making decisions. For example if a customer submits a French drivers licence here are some of the decisions we will need to make -

- Is the customer's business located in France? If not then we fail
- Is the expiry date < today ? If not then we fail
- Is the expiry date not on the front at line 4b ? If not then we fail

Naive Bayes does not support the concept of decision making that our solution requires

Decision Trees

This algorithm is for supervised learning and has the following attributes -

- Can be used for prediction of categories or regression.
- Commonly used for approval processing and fraud detection
- Training involves asking a series of questions on labelled input data. Therefore the logic inferred from our labelled input data will be used to make a prediction.
- Has proven to be successful with very large datasets[6]

The decision tree algorithm is best aligned to the problem we want to solve

Recommendation

We recommend that for this solution we used the supervised learning decision tree algorithm. It aligns with all of the attributes of our problem to be solved as outlined above.

References

[1] https://stripe.com/blog/untapped-opportunity-at-the-bottom-of-your-customer-funnel
[2] https://stripe.com/ie
[3] https://www.redflagalert.com/articles/a-guide-to-kyc-regulations-in-2024
[4]https://www.centralbank.ie/regulation/anti-money-laundering-and-countering-the-financing-of-terrorism/regulatory-requirements-guidance
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